

Current status of ARYA Macromolecular Crystallography beamline at SOLARIS NSRC

Joanna Sławek
Joint MXCuBE-ISPb meeting, 20.05.2025

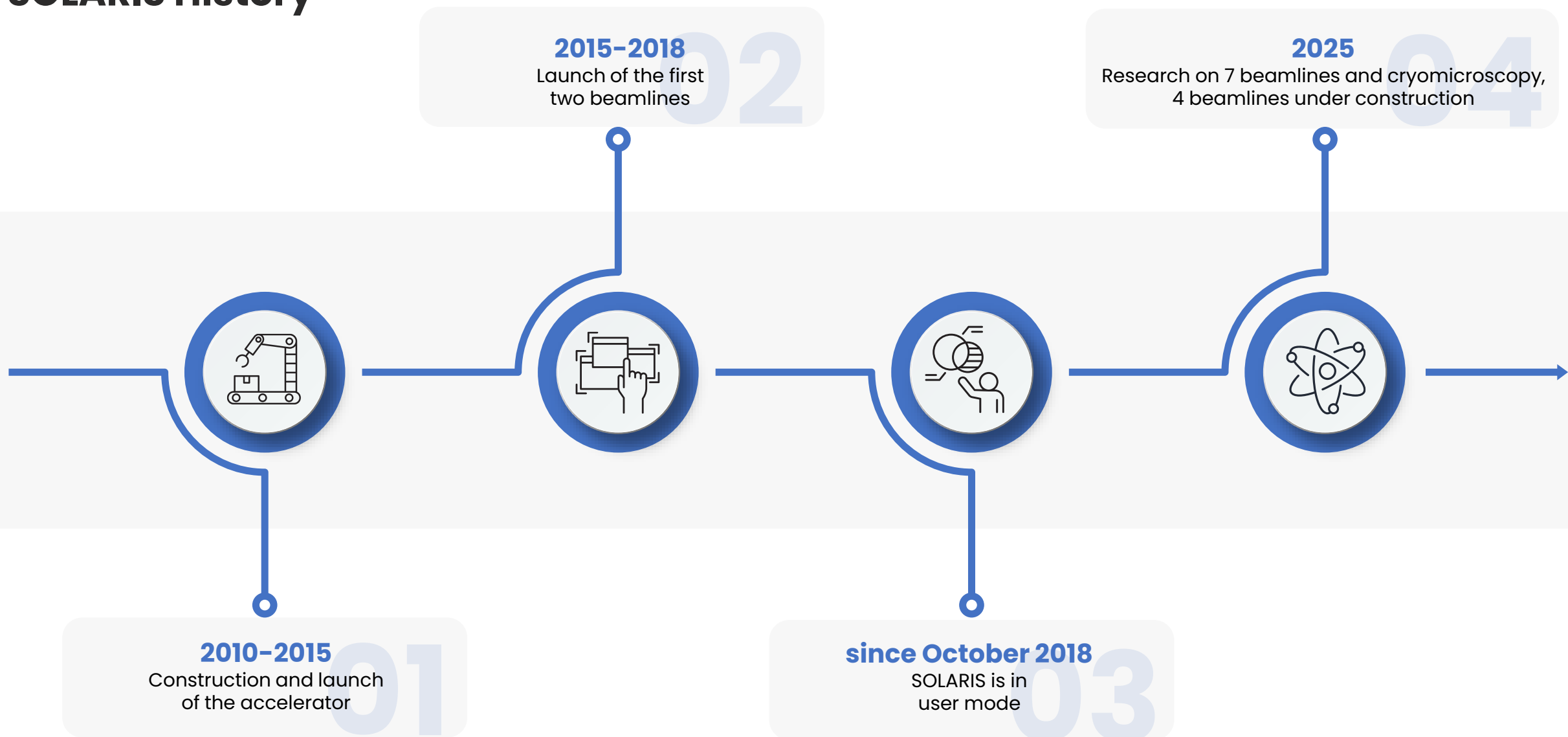
Introduction to SOLARIS

KRAKÓW
CITY CENTRE

NEW CAMPUS OF
THE JAGIELLONIAN UNIVERSITY

SOLARIS

SOLARIS History



Accelerators in SOLARIS

Basic data about the accelerator system

Circumference: 96 m

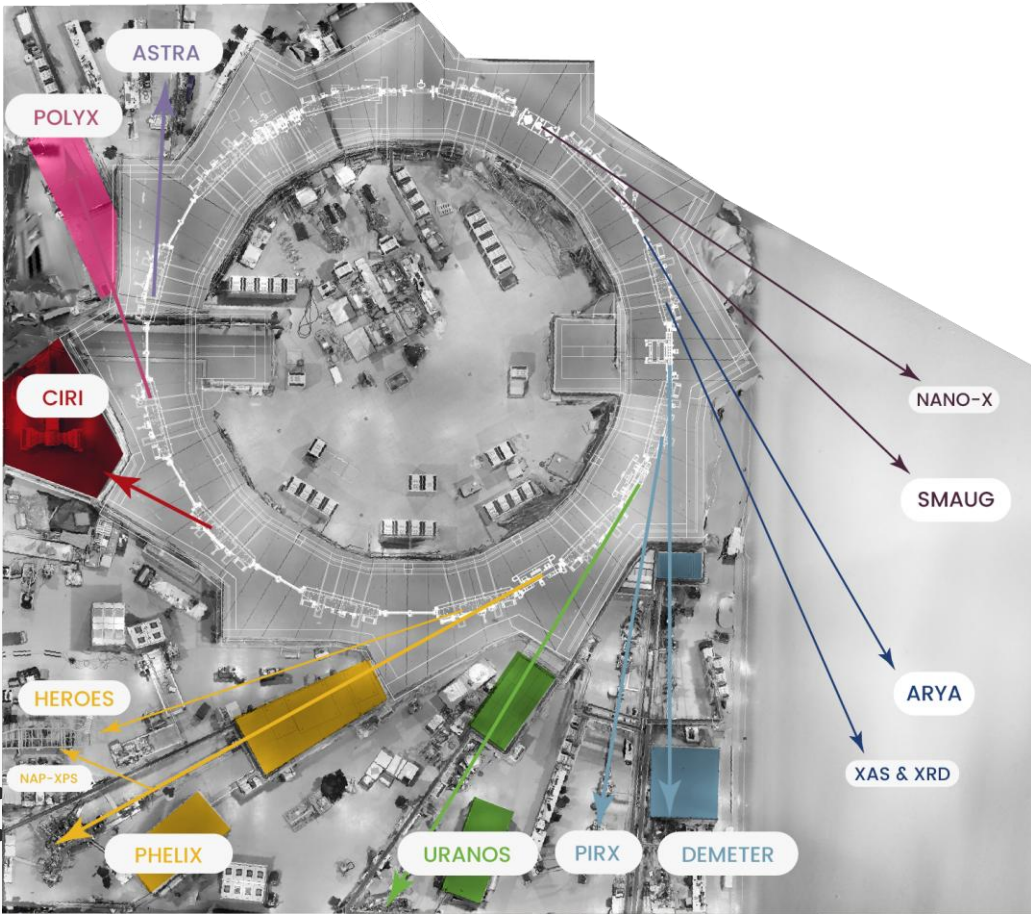
Energy: 1.5 GeV

12 bending magnets

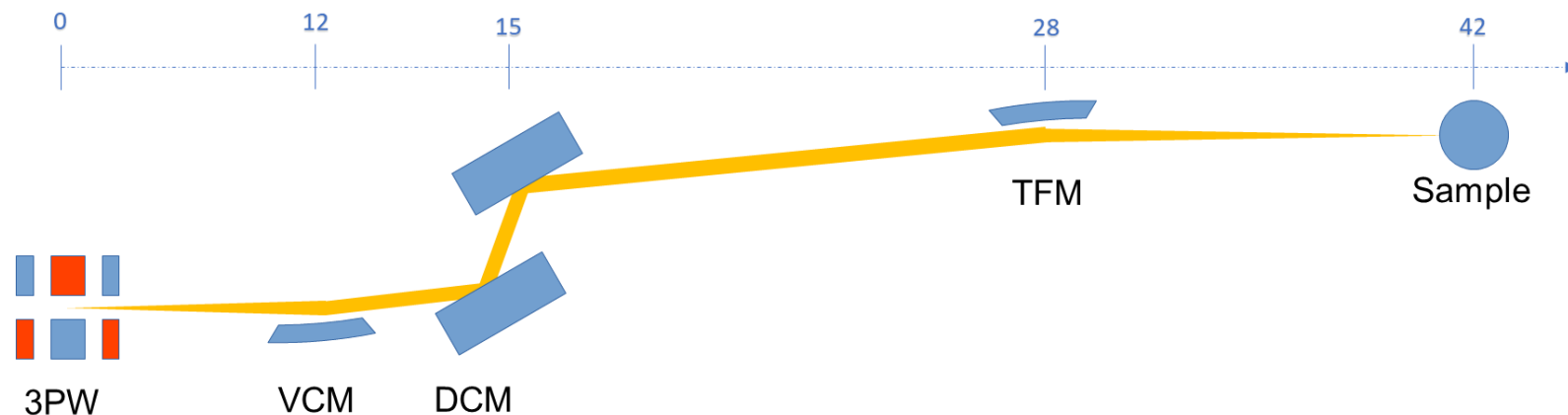


Research infrastructure

The Origins of the Names: SOLARIS and Its Beamlines



ARYA Beamline



- X-ray diffraction studies of proteins, nucleic acids and small molecules
- Crystallographic fragment screening
- High-pressure and low temperature data collection
- Fully automatic remote data collection
- Source: 3 pole-wiggler
- Wavelength tunability in a range 5–21 keV
- Photon flux at a sample $\sim 10^{11}$ – 10^{12} ph/s

TEAM



Joanna Sławek
ARYA beamline supervisor



Michał Gucwa
ARYA beamline scientist

CONSULTANTS



Tomasz Kołodziej
ARYA beamline consultant



Maciej Kozak
ARYA beamline consultant

PhD STUDENTS



Agnieszka Klonecka
PhD student



Jan Konopka
PhD student

SOLCRYS project

2015

First agreement between JU, AMU and PAS

2018

Agreement between JU and JINR

2019

SOLCRYS beamline project (MX and SAXS)

2020

Design and construction of SCW

2022

Withdrawal from the agreement with JINR and Budker

2023

MEN infrastructure grants for MX (II stage) and SAXS

2024

Preparation for tenders for I and II stage of the project

2025

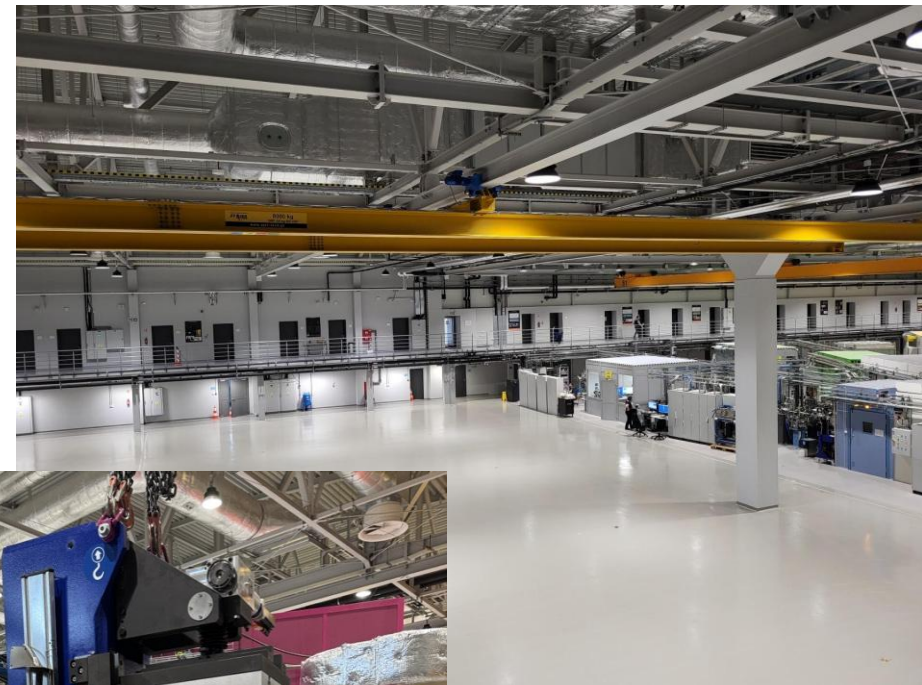
Tenders, Infrastructure project

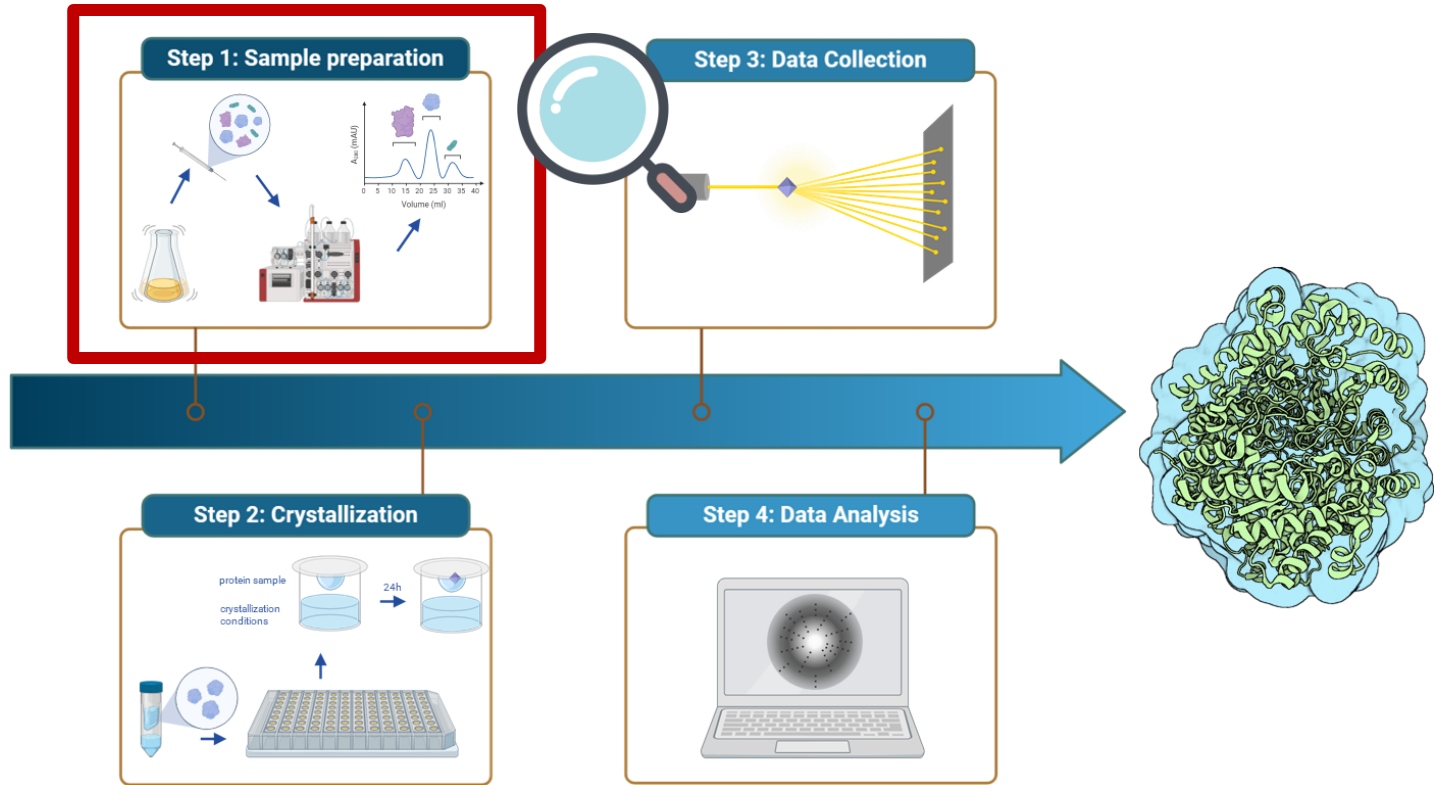
2026

Installation of endstation

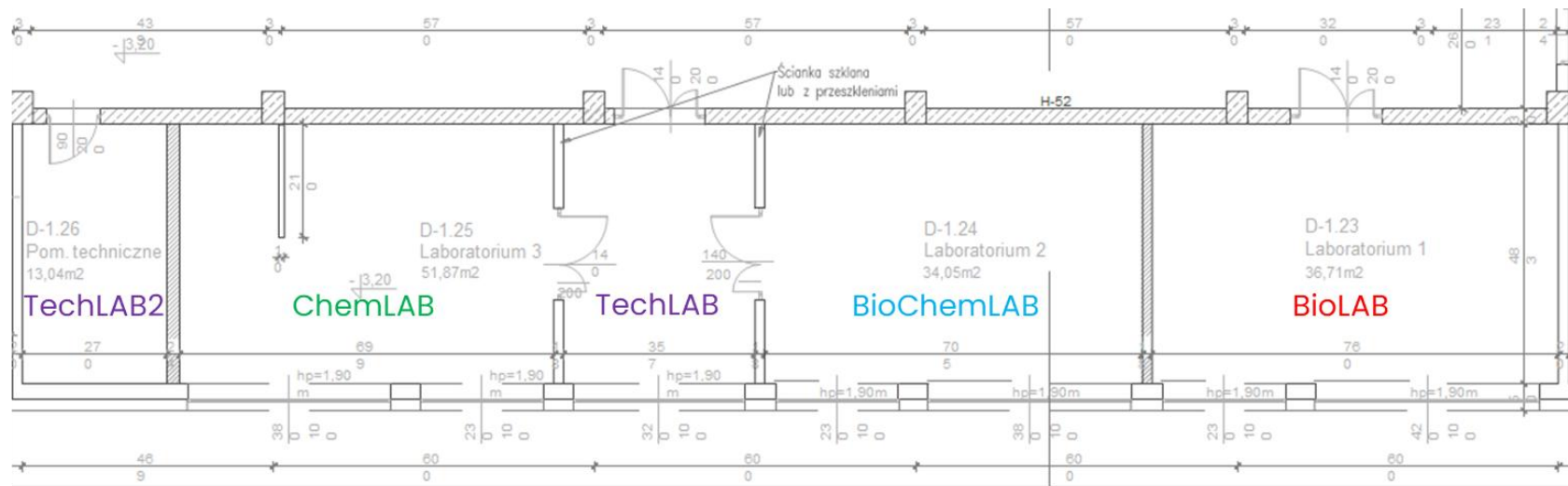
2027

Installation of optics, integration, comissioning

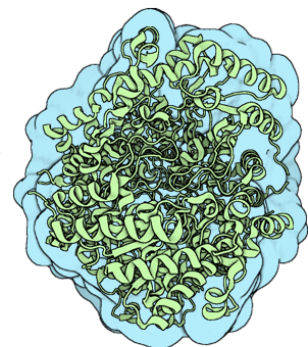
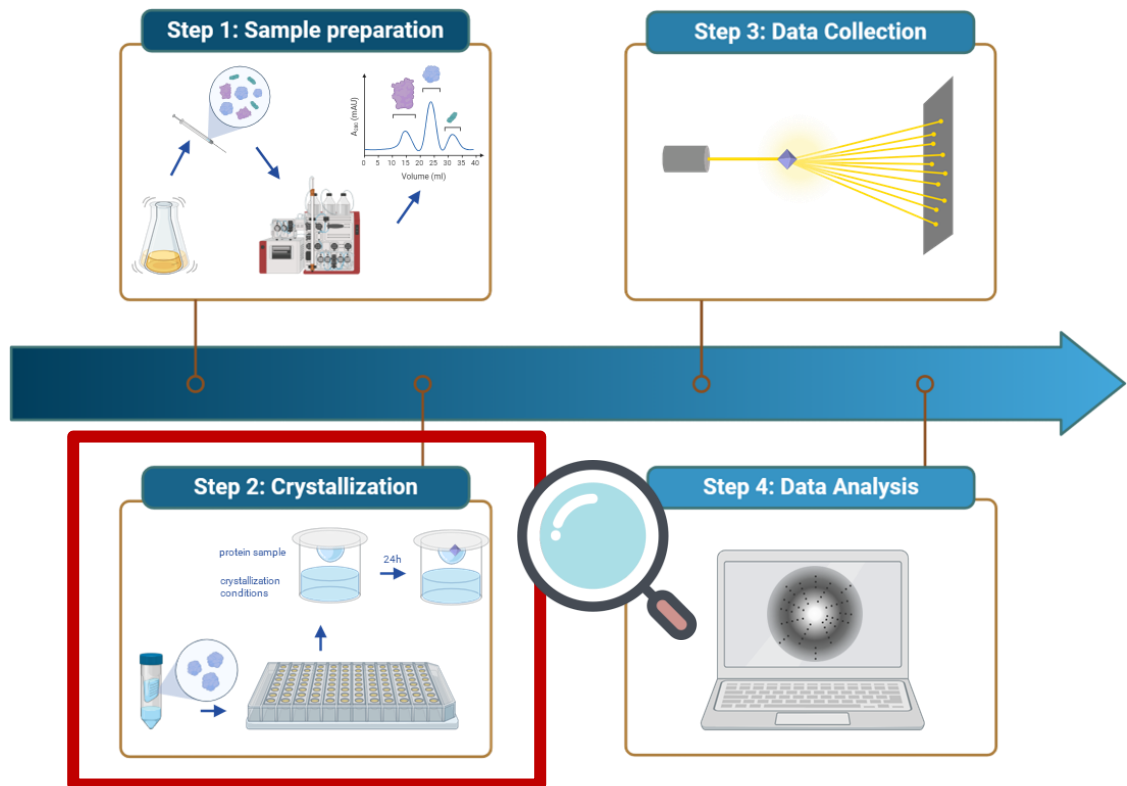




- Expression and purification of recombinant proteins
- Sample quality check
- Simple biochemistry



Labs will be operational in 2025



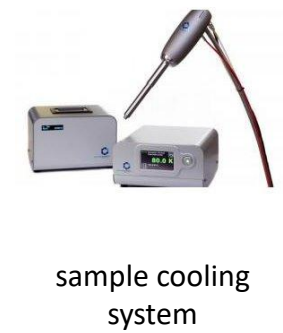
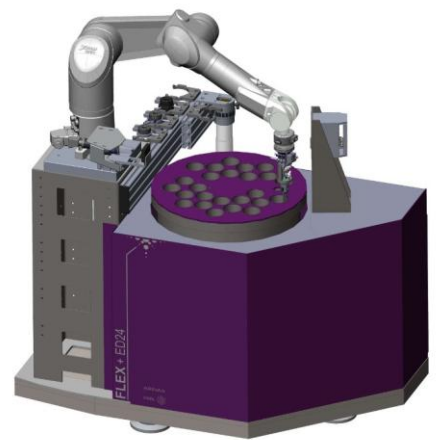
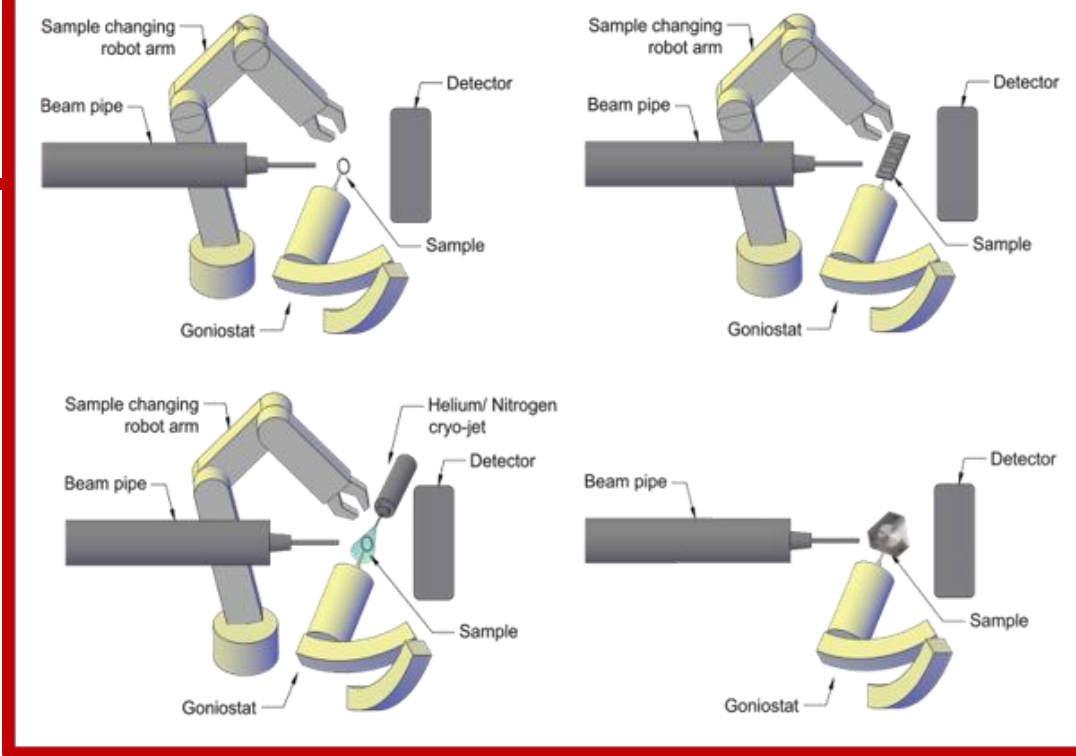
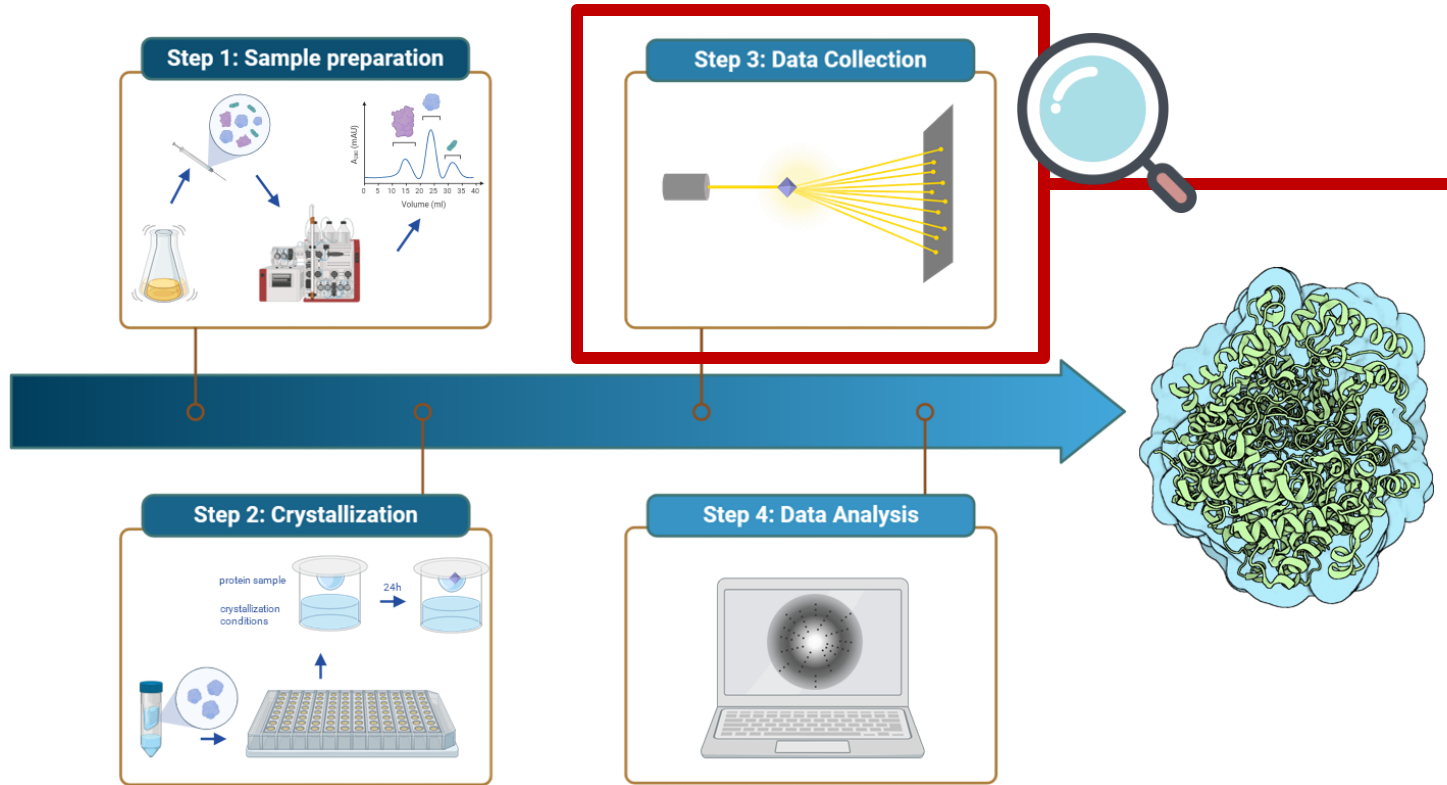
Screen builder

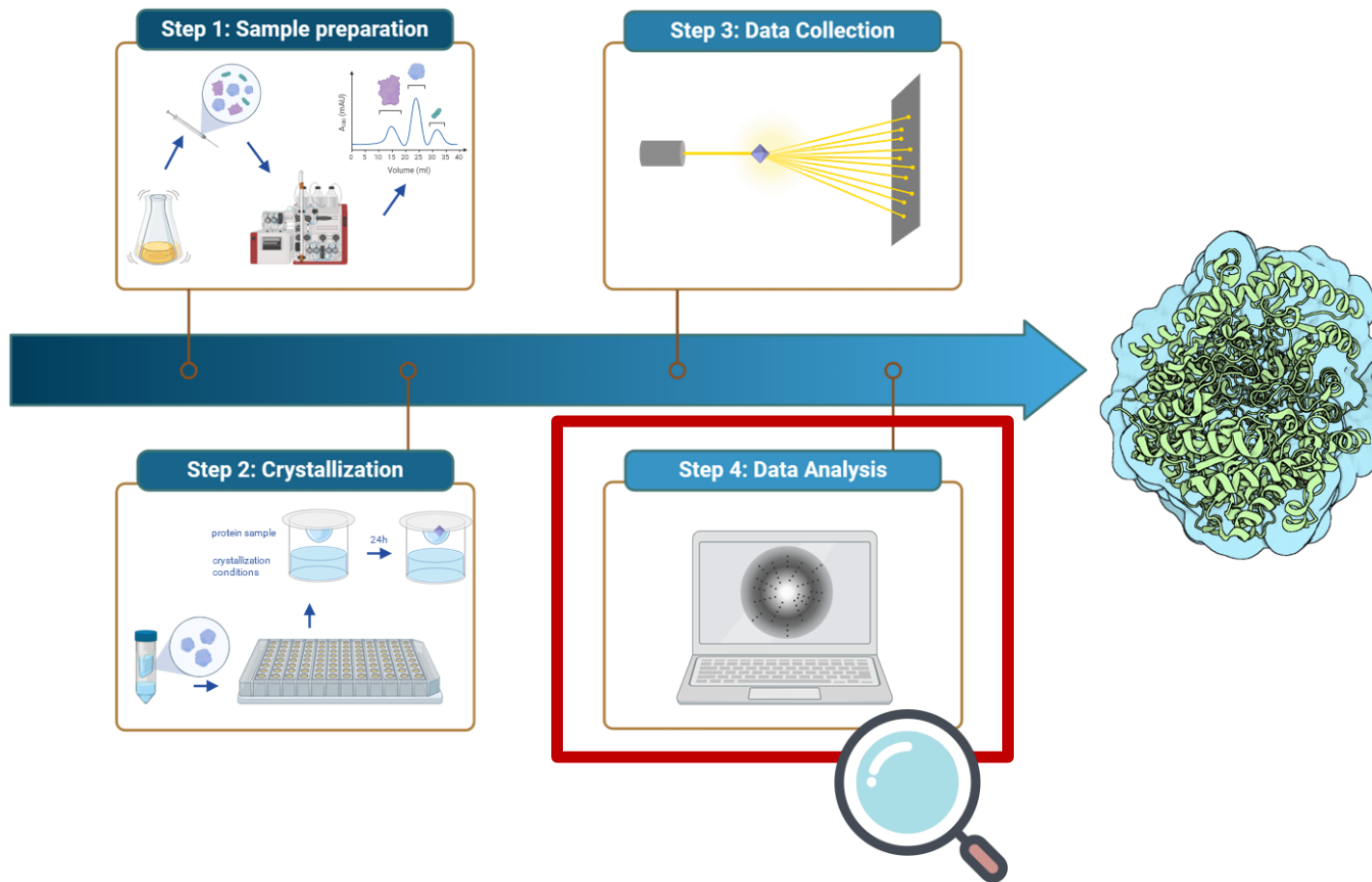


Crystallization robot



Plate storage with
crystal imager





Software:

- MXCuBE + ISpyB
- Commercial software packages for data analysis



**Any recommendations and
tips highly appreciated**